

Class Size: Courage — Answers

Courage

The abstract data generating mechanism coming out of Justice is:

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_n X_n + \epsilon, \quad \epsilon \sim N(0, \sigma^2)$$

Candidate Models

Create a candidate model, modeling the outcome `g4math` as a linear function of the class type `kind`. Call it `fit_kind`.

```
fit_kind <- linear_reg() |>
  fit(g4math ~ kind, data = star)

fit_kind |>
  tidy(conf.int = TRUE) |>
  select(term, estimate, conf.low, conf.high) |>
  mutate(across(where(is.numeric), \(x) round(x, 2)))
```

```
# A tibble: 3 x 4
  term                estimate conf.low conf.high
<chr>                <dbl>    <dbl>    <dbl>
1 (Intercept)         710.      707.      712.
2 kindregular with aid  -1.89    -6.04     2.27
3 kindsmall            -0.34    -4.6      3.92
```

Both class-type coefficients have confidence intervals comfortably including zero — class type alone tells us essentially nothing about fourth-grade math scores, matching what the EDA showed. Perhaps class type matters once we account for who is in each class, so we add `race`.

Create a candidate model, modeling `g4math` as a linear function of both `kind` and `race`. Call it `fit_kind_race`.

```
fit_kind_race <- linear_reg() |>
  fit(g4math ~ kind + race, data = star)

fit_kind_race |>
  tidy(conf.int = TRUE) |>
  select(term, estimate, conf.low, conf.high) |>
  mutate(across(where(is.numeric), \(x) round(x, 2)))
```

```
# A tibble: 5 x 4
```

	term <chr>	estimate <dbl>	conf.low <dbl>	conf.high <dbl>
1	(Intercept)	696.	691.	701.
2	kindregular with aid	-2	-6.12	2.12
3	kindsmall	-0.22	-4.44	4
4	raceOthers	38.3	12.6	63.9
5	raceWhite	16.1	11.2	20.9

Race carries real signal — White students score about 16 points higher than Black students, with an interval far from zero — but the class-type coefficients barely move and still include zero. One more candidate: class type records only where a student sat in one year, while `yearssmall` records the *dose* — how many years (0–4) the student spent in small classes.

Create a candidate model which adds `yearssmall` to `fit_kind_race`. Call it `fit_kind_race_years`.

```
fit_kind_race_years <- linear_reg() |>
  fit(g4math ~ kind + race + yearssmall, data = star)

fit_kind_race_years |>
  tidy(conf.int = TRUE) |>
  select(term, estimate, conf.low, conf.high) |>
  mutate(across(where(is.numeric), \(x) round(x, 2)))
```

```
# A tibble: 6 x 4
```

	term <chr>	estimate <dbl>	conf.low <dbl>	conf.high <dbl>
1	(Intercept)	695.	690.	700.
2	kindregular with aid	-1.95	-6.07	2.17
3	kindsmall	-7.66	-15.9	0.61
4	raceOthers	38.7	13.1	64.3

5	raceWhite	16.2	11.4	21.1
6	yearssmall	2.18	0.1	4.27

Now something interesting appears: each additional year spent in a small class is worth about 2.2 points, with an interval excluding zero. The snapshot (`kind`) shows nothing; the dose (`yearssmall`) shows a small but real association. Note that `kindsmall` turns negative (-7.66) once `yearssmall` enters — the two variables are entangled, since students assigned to small classes accumulate more years in them; a student assigned small who stays all four years nets $-7.66 + 4 \times 2.18 = +1$ point relative to a regular-class student.

The Data Generating Mechanism

Which model should we choose, and why? Define `fit_classsize` as your chosen model and write out the fitted equation.

```
fit_classsize <- linear_reg() |>
  fit(g4math ~ kind + race + yearssmall, data = star)

fit_classsize |>
  tidy(conf.int = TRUE) |>
  select(term, estimate, conf.low, conf.high) |>
  mutate(across(where(is.numeric), \(x) round(x, 2)))
```

```
# A tibble: 6 x 4
  term                estimate conf.low conf.high
  <chr>                <dbl>    <dbl>    <dbl>
1 (Intercept)         695.      690.      700.
2 kindregular with aid  -1.95    -6.07     2.17
3 kindsmall            -7.66   -15.9     0.61
4 raceOthers           38.7     13.1     64.3
5 raceWhite            16.2     11.4     21.1
6 yearssmall           2.18      0.1      4.27
```

We choose the full model. `race` and `yearssmall` earn their places with intervals excluding zero. The `kind` coefficients do not — but `kind` is the treatment the causal question is about, so the question, not the data, forces it into the model. Reporting “we cannot distinguish the assignment effect from zero” requires keeping the variable whose effect we cannot distinguish.

$$\widehat{g4math} = 695 - 1.95 \cdot \text{regular_with_aid} - 7.66 \cdot \text{small} + 38.7 \cdot \text{Others} + 16.2 \cdot \text{White} + 2.18 \cdot \text{yearssmall}$$

The intercept describes a Black student in a regular class with zero years in small classes. **This is our data generating mechanism.**

Model Checking

Use `check_predictions(extract_fit_engine(fit_classsize))` to perform a posterior predictive check of your chosen model.

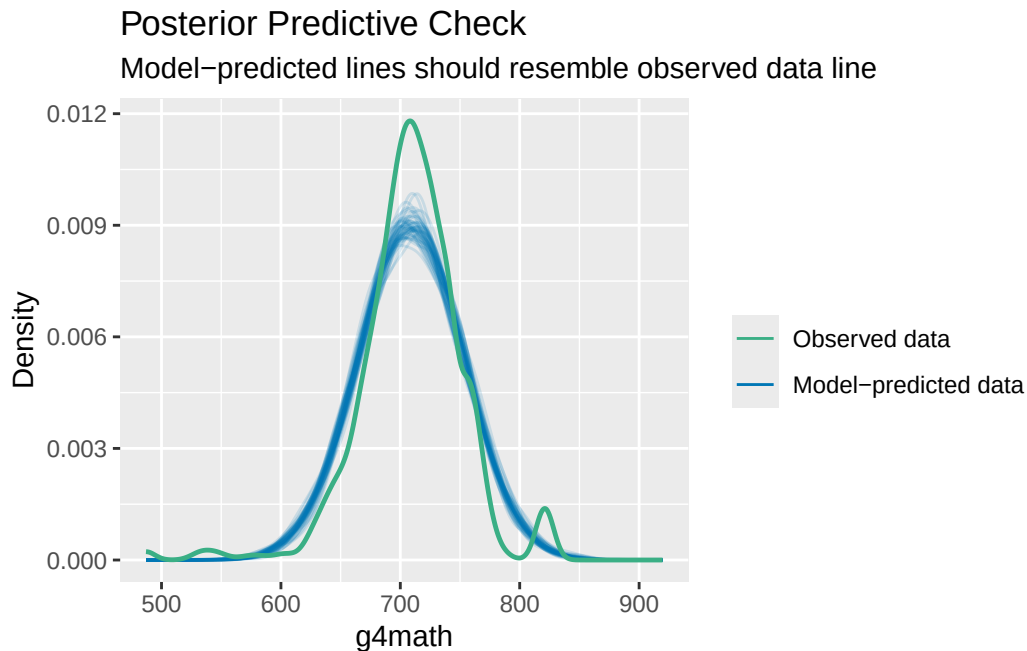
```
check_predictions(extract_fit_engine(fit_classsize))
```

Warning: Minimum value of original data is not included in the replicated data.

Model may not capture the variation of the data.

Ignoring unknown labels:

```
* size : ""
```



`check_predictions()` simulates new math scores from the fitted model and overlays them on the observed distribution. The simulated curves track the observed scores well in the center, so the model is not obviously misdescribing the outcome — the check we want to pass before trusting the predictions in Temperance.